

Relations between equivalences of bent functions

Paul Leopardi

Australian National University (professional staff)
ACCESS-NRI

For Australian Mathematical Society Conference 2023

Motivation

Question:

Which strongly regular graphs arise as Cayley graphs of bent Boolean functions?

Bent functions

Definition 1

The Walsh Hadamard transform of a Boolean function $f : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$ is

$$W_f(x) := \sum_{y \in \mathbb{F}_2^{2m}} (-1)^{f(y) + \langle x, y \rangle}$$

Definition 2

*A Boolean function $f : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$ is **bent** if and only if its Walsh Hadamard transform has constant absolute value 2^m .*

(Dillon 1974; Rothaus 1976)

The Cayley graph of a Boolean function

Definition 3

The *Cayley graph* $\text{Cay}(f)$ of a Boolean function

$$f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2 \quad \text{where} \quad f(0) = 0$$

is an undirected graph with

$$V(\text{Cay}(f)) := \mathbb{F}_2^n, \quad (x, y) \in E(\text{Cay}(f)) \Leftrightarrow f(x + y) = 1.$$

Strongly regular graphs

Definition 4

A simple graph Γ of order v is *strongly regular* with parameters (v, k, λ, μ) if

- ▶ each vertex has degree k ,
- ▶ each adjacent pair of vertices has λ common neighbours, and
- ▶ each nonadjacent pair of vertices has μ common neighbours.

(Brouwer, Cohen and Neumaier 1989)

Cayley graphs of bent functions

Proposition 1

(Bernasconi and Codenotti 1999)

The Cayley graph $\text{Cay}(f)$ of a bent function f on $\mathbb{F}_2^{2m}$ with $f(0) = 0$ is a strongly regular graph with $\lambda = \mu$.

The parameters of $\text{Cay}(f)$ are

$$(v, k, \lambda) = (4^m, 2^{2m-1} - 2^{m-1}, 2^{2m-2} - 2^{m-1})$$

$$\text{or } (4^m, 2^{2m-1} + 2^{m-1}, 2^{2m-2} + 2^{m-1}).$$

(Bernasconi and Codenotti 1999)

Extended affine equivalence

Definition 5

For bent functions $f, g : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$,

f is *extended affine equivalent* to g if and only if

$$g(x) = f(Ax + b) + \langle c, x \rangle + \delta$$

for some $A \in GL(2m, 2)$, $b, c \in \mathbb{F}_2^{2m}$, $\delta \in \mathbb{F}_2$.

(Budaghyan, Carlet and Pott 2006)

General linear equivalence

Definition 6

For bent functions $f, g : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$, f is *general linear equivalent* to g if and only if

$$g(x) = f(Ax)$$

for some $A \in GL(2m, 2)$.

Extended translation equivalence

Definition 7

For bent functions $f, g : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$,
 f is *extended translation equivalent* to g if and only if

$$g(x) = f(x + b) + \langle c, x \rangle + \delta$$

for $b, c \in \mathbb{F}_2^{2m}$, $\delta \in \mathbb{F}_2$.

Cayley equivalence

Definition 8

For $f, g : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$, with both f and g bent,
we call f and g **Cayley equivalent**, and write $f \equiv g$,
if and only if $f(0) = g(0) = 0$ and $\text{Cay}(f) \equiv \text{Cay}(g)$ as graphs.

Equivalently, $f \equiv g$ if and only if $f(0) = g(0) = 0$ and
there exists a bijection $\pi : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2^{2m}$ such that

$$g(x + y) = f(\pi(x) + \pi(y)) \quad \text{for all } x, y \in \mathbb{F}_2^{2m}.$$

Extended Cayley equivalence

Definition 9

For $f, g : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$, with both f and g bent,
if there exist $\delta, \epsilon \in \{0, 1\}$ such that $f + \delta \equiv g + \epsilon$,
we call f and g **extended Cayley (EC) equivalent** and write $f \cong g$.

Extended Cayley equivalence is an equivalence relation on the set of all bent functions on $\mathbb{F}_2^{2m}$.

General linear equivalence implies Cayley equivalence

Theorem 1

If f is bent with $f(0) = 0$ and $g(x) := f(Ax)$ where $A \in GL(2m, 2)$, then g is bent with $g(0) = 0$ and $f \equiv g$.

Proof.

$$g(x + y) = f(A(x + y)) = f(Ax + Ay) \quad \text{for all } x, y \in \mathbb{F}_2^{2m}.$$



Extended affine, extended translation, and extended Cayley equivalence (1)

Theorem 2

For $A \in GL(2m, 2)$, $b, c \in \mathbb{F}_2^{2m}$, $\delta \in \mathbb{F}_2$, $f : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$,
the function

$$h(x) := f(Ax + b) + \langle c, x \rangle + \delta$$

can be expressed as $h(x) = g(Ax)$ where

$$g(x) := f(x + b) + \langle (A^{-1})^T c, x \rangle + \delta,$$

and therefore if f is bent then $h \cong g$.

Extended affine, extended translation, and extended Cayley equivalence (2)

Therefore, to determine which extended Cayley equivalence classes have members within the extended affine equivalence class of a bent function $f : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$ (for which $f(0) = 0$) we need only examine the extended translation equivalent functions of the form

$$f(x + b) + \langle c, x \rangle + f(b),$$

for each $b, c \in \mathbb{F}_2^{2m}$.

Weights and weight classes

Definition 10

The *weight* of a binary function is the cardinality of its *support*.
For f on $\mathbb{F}_2^{2m}$

$$\text{supp}(f) := \{x \in \mathbb{F}_2^{2m} \mid f(x) = 1\}.$$

A bent function f on $\mathbb{F}_2^{2m}$ has weight

$$\text{wt}(f) = 2^{2m-1} - 2^{m-1} \quad (\text{weight class } \text{wc}(f) = 0), \text{ or}$$

$$\text{wt}(f) = 2^{2m-1} + 2^{m-1} \quad (\text{weight class } \text{wc}(f) = 1).$$

Quadratic bent functions have two General Linear classes

Theorem 3

For each $m > 0$, the extended affine equivalence class of quadratic bent functions $q : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$ contains members of exactly two General linear equivalence classes, corresponding to the two possible weight classes of $x \mapsto q(x + b) + \langle c, x \rangle + q(b)$.

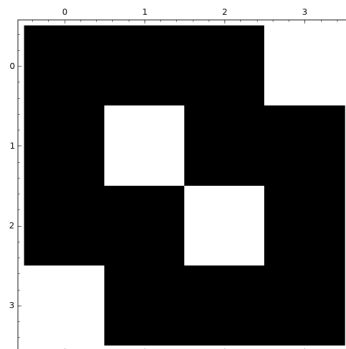
Quadratic bent functions have two extended Cayley classes

Corollary 4

For each $m > 0$, the extended affine equivalence class of quadratic bent functions $q : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$ contains exactly two extended Cayley equivalence classes, corresponding to the two possible weight classes of $x \mapsto q(x + b) + \langle c, x \rangle + q(b)$.

For 2 dimensions: ET class $[f_{2,1}]$

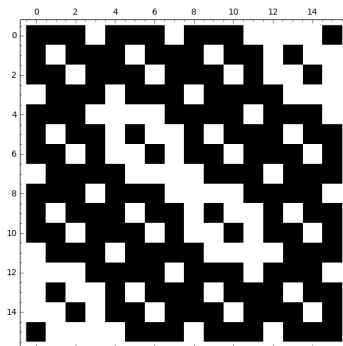
One extended affine class, containing the extended translation class $[f_{2,1}]$, where $f_{2,1}(x) := x_0x_1$.



$[f_{2,1}]$: 2 extended Cayley classes, 2
General Linear classes

For 4 dimensions: ET class $[f_{4,1}]$

One extended affine class, containing the extended translation class $[f_{4,1}]$, where $f_{4,1}(x) := x_0x_1 + x_2x_3$.



$[f_{4,1}]$: 2 extended Cayley classes, 2
General Linear classes

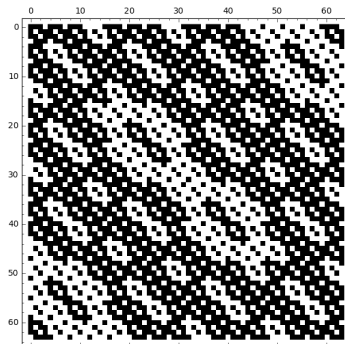
For 6 dimensions: ET classes

Four extended affine classes, containing the following extended translation classes:

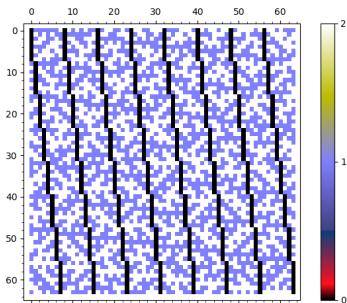
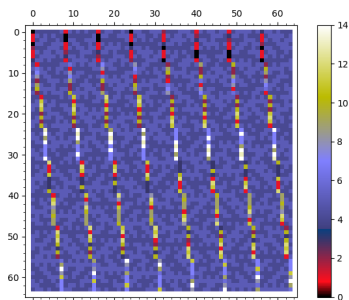
Class	Representative
$[f_{6,1}]$	$f_{6,1} := x_0x_1 + x_2x_3 + x_4x_5$
$[f_{6,2}]$	$f_{6,2} := x_0x_1x_2 + x_0x_3 + x_1x_4 + x_2x_5$
$[f_{6,3}]$	$f_{6,3} := x_0x_1x_2 + x_0x_1 + x_0x_3 + x_1x_3x_4 + x_1x_5$ $+ x_2x_4 + x_3x_4$
$[f_{6,4}]$	$f_{6,4} := x_0x_1x_2 + x_0x_3 + x_1x_3x_4 + x_1x_5 + x_2x_3x_5$ $+ x_2x_3 + x_2x_4 + x_2x_5 + x_3x_4 + x_3x_5$

(Rothaus 1976; Tokareva 2015)

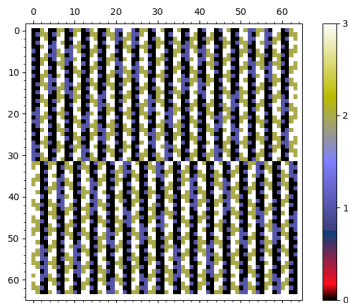
ET class $[f_{6,1}]$



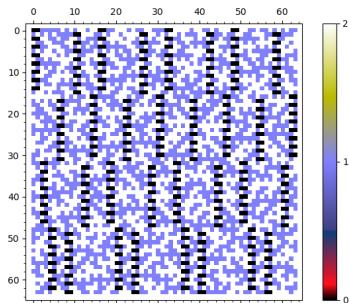
$[f_{6,1}]$: 2 extended Cayley classes,
2 General Linear classes

ET class $[f_{6,2}]$  $[f_{6,2}]$: 3 extended Cayley classes $[f_{6,2}]$: 15 General Linear classes

Since $f_{6,1} \equiv f_{6,2}$, the Cayley graph for extended Cayley class 0 is isomorphic to the Cayley graph for class 0 of $[f_{6,1}]$.

ET classes $[f_{6,3}]$ and $[f_{6,4}]$ 

$[f_{6,3}]$: 4 extended Cayley classes,
4 General Linear classes



$[f_{6,4}]$: 3 extended Cayley classes,
3 General Linear classes

Preprint, source code and documentation

Preprint: Paul Leopardi, Classifying bent functions by their Cayley graphs, arXiv:1705.04507 [math.CO]. Revised, December, 2023.

CoCalc: Public worksheets, Sage and Python source code

<http://tinyurl.com/Boolean-Cayley-graphs>

GitHub: Sage and Python source code

<https://github.com/penguian/Boolean-Cayley-graphs>

SourceForge: Documentation

<https://boolean-cayley-graphs.sourceforge.io/>

References (1)

1. A. Bernasconi and B. Codenotti. Spectral analysis of Boolean functions as a graph eigenvalue problem. *IEEE Transactions on Computers*, 48(3):345–351, (1999).
2. A. Braeken. *Cryptographic Properties of Boolean Functions and S-Boxes*. PhD thesis, Katholieke Universiteit Leuven, Belgium, (2006).
3. A. Brouwer, A. Cohen, and A. Neumaier. *Distance-Regular Graphs*. *Ergebnisse der Mathematik und Ihrer Grenzgebiete*, vol. 18. Berlin, Heidelberg, [Germany] : Springer-Verlag, (1989).
4. L. Budaghyan, C. Carlet, and A. Pott. New classes of almost bent and almost perfect nonlinear polynomials. *IEEE Transactions on Information Theory*, 6;52(3):1141-52 (2006).

References (2)

5. J. F. Dillon. *Elementary Hadamard Difference Sets*. PhD thesis, University of Maryland College Park, Ann Arbor, USA, (1974).
6. O. S. Rothaus. On “bent” functions. *Journal of Combinatorial Theory, Series A*, 20(3):300–305, (1976).
7. N. Tokareva. *Bent functions: results and applications to cryptography*. Academic Press, (2015).

Acknowledgements

Robin Bowen, An Braeken, Nathan Clisby, Robert Craigen, Joanne Hall, David Joyner, Philippe Langevin, Matthew Leingang, William Martin, Padraig Ó Catháin, Judy-anne Osborn, Dima Pasechnik, William Stein, Natalia Tokareva, and Sanming Zhou.

Australian National University. University of Newcastle, Australia. University of Melbourne. Australian Government - Bureau of Meteorology. National Computational Infrastructure. ACCESS-NRI.

SageMath, CoCalc, Bliss, Nauty, MPI4py, SQLite3, DB Browser for SQLite, PostgreSQL, Psycopg2.